Demonstration of Proposed Features

Date	06 July 2026
Team ID	Not applicable / not assigned
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S. No	Feature Name	Description	Status	Demonstrated	Remarks
1	Data Preprocessing & Feature Engineering	Merge application_record.csv and credit_record.csv, handle missing values/duplicates, and label-encode categorical fields (gender, income type, education, family status, etc.)	Implemented	Yes	Verified in main.py; runs cleanly end-to-end on both CSV files
2	Target Label Engineering from Credit History	Derive a Good/Bad customer label per applicant from monthly STATUS codes in credit_record.csv, aggregated per customer ID	Implemented	Yes	STATUS values mapped to a binary TARGET and grouped by ID
3	Multi-Model Training & Comparison	Train and evaluate Logistic Regression, Decision Tree, and Random Forest classifiers on Accuracy and F1-score	Implemented	Yes	Confusion matrix and classification report printed for all three models
4	Model Persistence	Serialize the best-performing model (Random Forest) to model.pkl for reuse by the web app	Implemented	Yes	model.pkl present in repository and loaded successfully by app.py
5	Flask Web Application for Real-Time Prediction	Web form to collect applicant details and return an instant Good Customer / Bad Customer prediction	Implemented	Yes	Tested locally via python app.py at http://127.0.0.1:5000
6	Public Deployment (Render)	Host the Flask app publicly on Render so it can be accessed via a shareable URL	Pending	No	Not yet deployed; requirements.txt and production config still need to be added

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	83% (5 of 6 proposed features implemented and demonstrated)